MAE 155A: Aerospace Engineering Design I

Aircraft system design

Winter 2026

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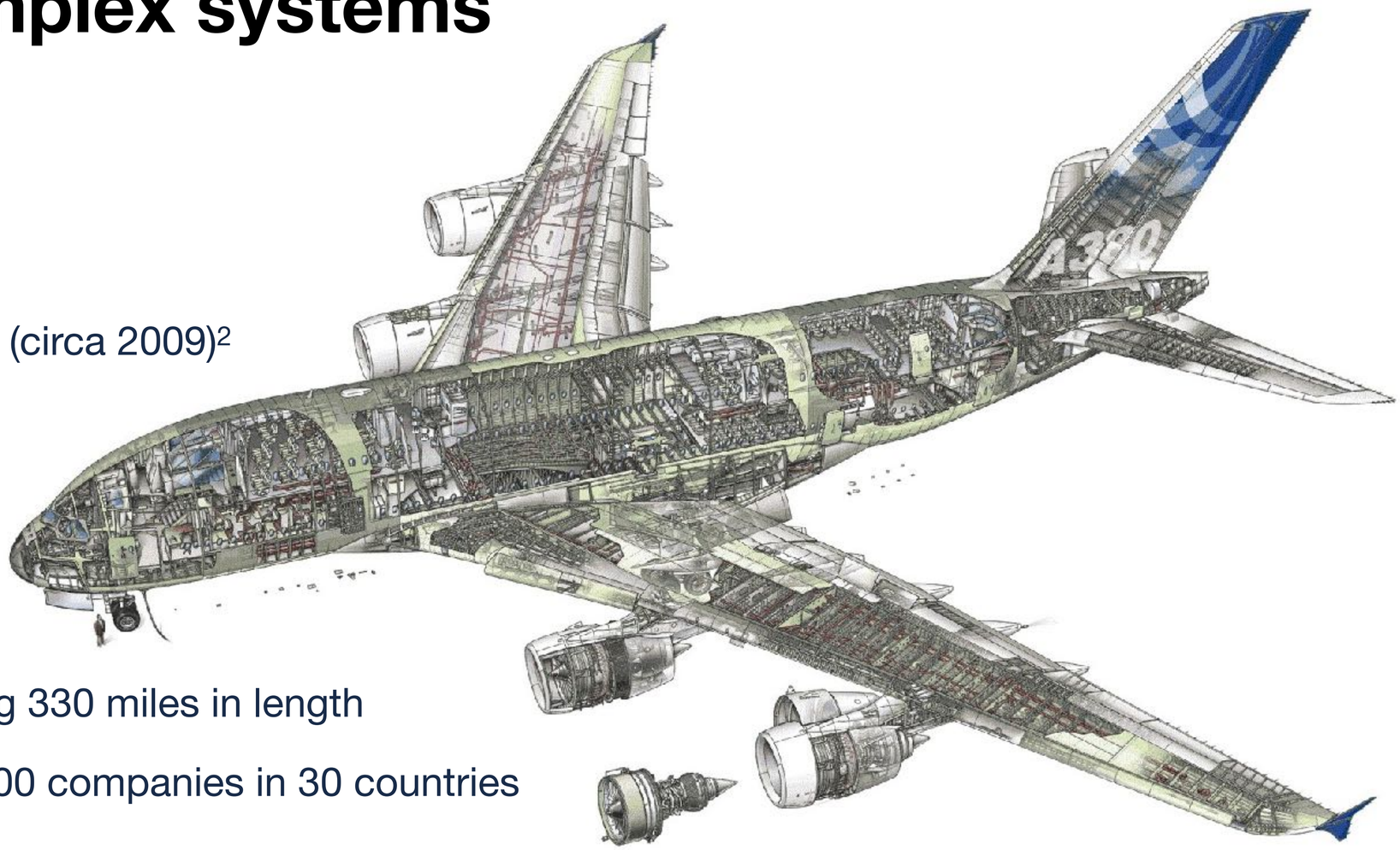
Mechanical and Aerospace Engineering

Aircraft are complex systems



Unit cost ~$430 million (2016)[1]

Development cost ~$25 billion (circa 2009)[2]

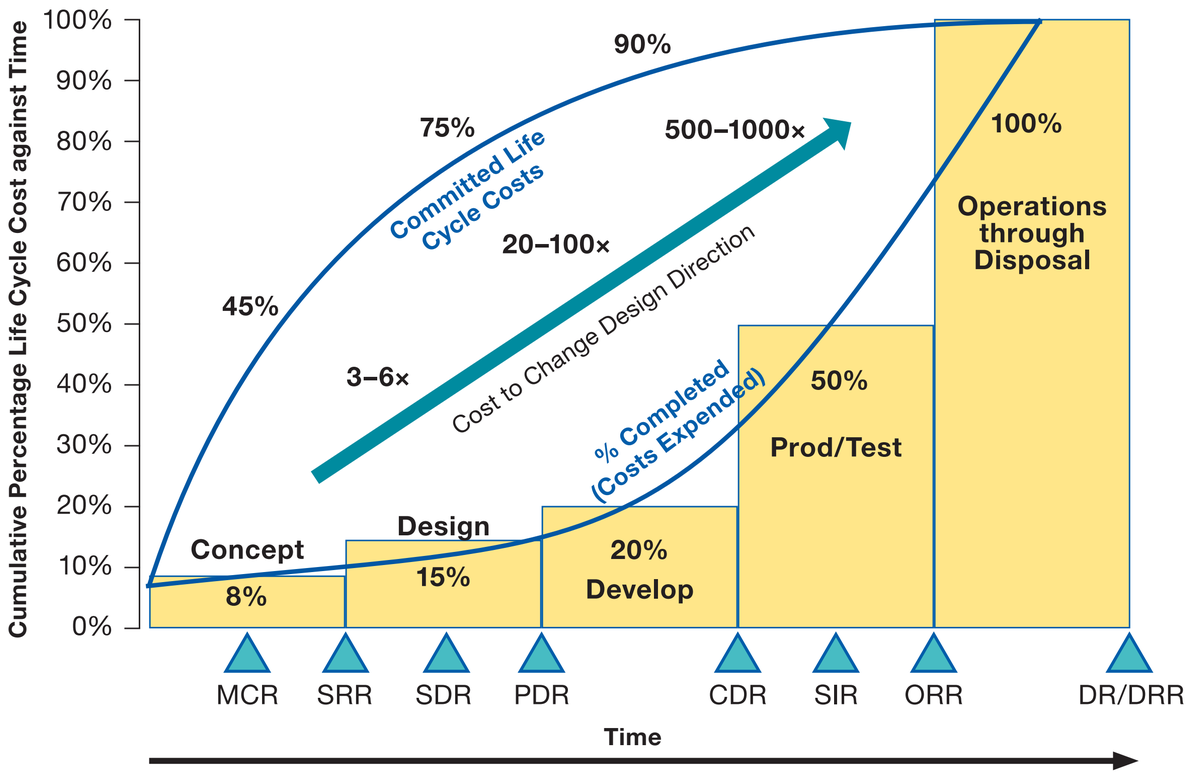
3~4 million parts

100,000 different wires, totaling 330 miles in length

Components produced by 1,500 companies in 30 countries

[1] Airbus Press Office. (2016). Airbus Aircraft 2016 Average List Prices.
[2] Johnsen, T. E., & Lewis, M. (2009, April). Supplier involvement in the development of the A380 super jumbo. In Proceedings of the 17th Annual IPSERA Conference.

Early decisions have disproportionately high impact on lifecycle costs



MCR	Mission Concept Review	**CDR**	Critical Design Review
SRR	System Requirements Review	**SIR**	System Integration Review
SDR	System Definition Review	**ORR**	Operational Readiness Review
PDR	Preliminary Design Review	**DR/DRR**	Decommissioning/Disposal Readiness Review

Hirshorn, S. R., Voss, L. D., & Bromley, L. K. (2017). NASA systems engineering handbook (No. HQ-E-DAA-TN38707). Chicago.

Agenda

System design

Aircraft design process

Conceptual design

Definitions

A **system** is "*the combination of elements that function together to produce the capability required to meet a need. The elements include all hardware, software, equipment, facilities, personnel, processes, and procedures needed for this purpose.*" [Hirshorn et al., 2017]

Systems engineering is "*a methodical, multi-disciplinary approach for the **design**, realization, technical management, operations, and retirement of a system.*" [Hirshorn et al., 2017]

- Scope: It aims to ensure that a system meets stakeholder needs and performs its intended function effectively while balancing technical, cost, and schedule constraints, considering the full lifecycle.

System design is a subset of systems engineering that is focused on the definition of the system's design, i.e., the architecture, components, and interfaces.

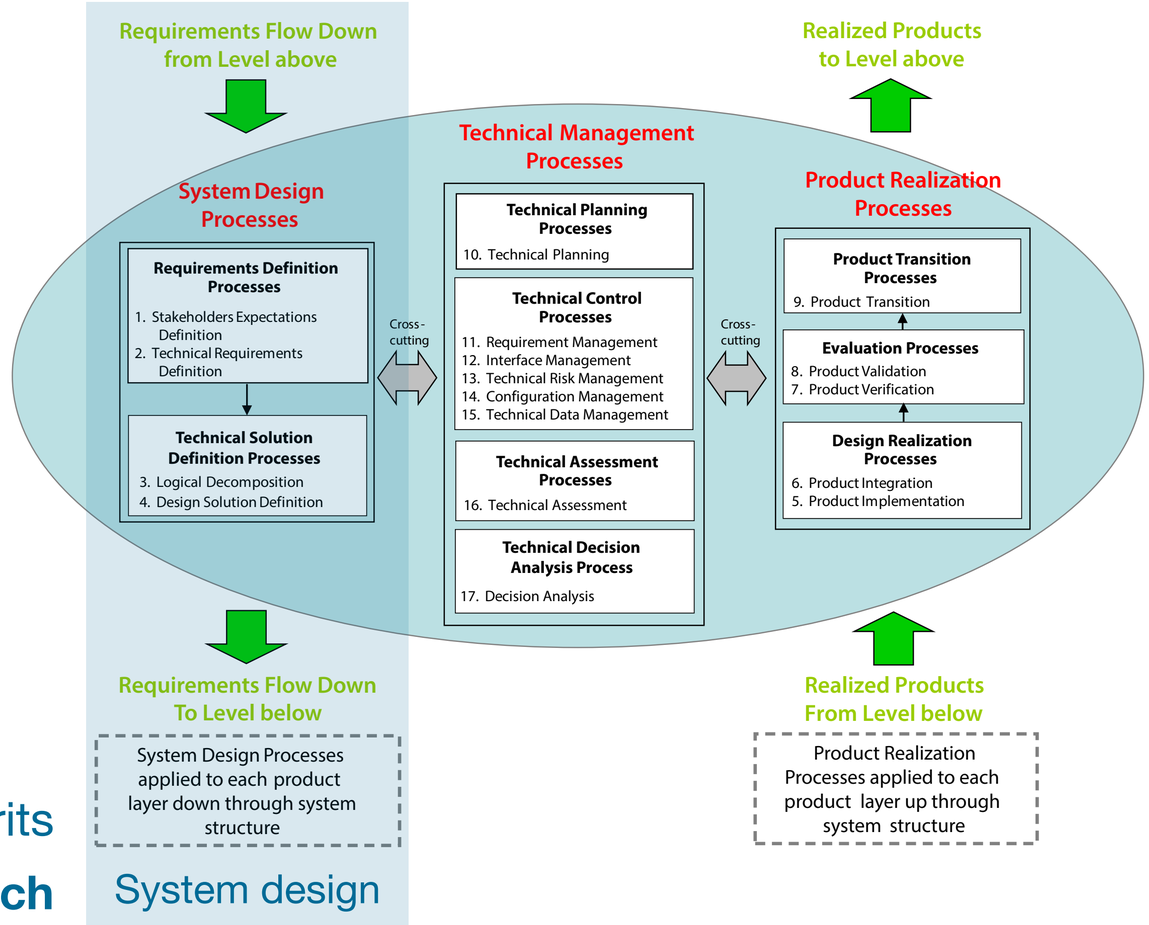
- Scope: It aims to describe and specify the system, rather than coordinate or manage its development or operation.

Hirshorn, S. R., Voss, L. D., & Bromley, L. K. (2017). NASA systems engineering handbook (No. HQ-E-DAA-TN38707). Chicago.

Systems engineering manages complexity using a top-down approach

System design process

- Start by making assumptions
- Rely on existing solutions (i.e., designs)
- Iterate and refine



System design inherits this **top-down approach**

Systems engineering considers the full lifecycle

Full lifecycle

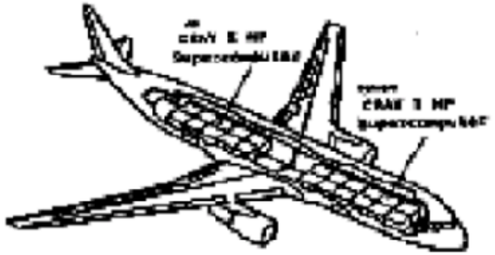
Phase		Purpose	Typical Outcomes
Pre-Formulation	**Pre-Phase A** Concept Studies	To produce a broad spectrum of ideas and alternatives for missions from which new programs/projects can be selected. Determine feasibility of desired system, develop mission concepts, draft system-level requirements, assess performance, cost, and schedule feasibility; identify potential technology needs, and scope.	Feasible system concepts in the form of simulations, analysis, study reports, models, and mock-ups
Formulation	**Phase A** Concept and Technology Development	To determine the feasibility and desirability of a suggested new system and establish an initial baseline compatibility with NASA's strategic plans. Develop final mission concept, system-level requirements, needed system technology developments, and program/project technical management plans.	System concept definition in the form of simulations, analysis, engineering models and mock-ups, and trade study definition
	Phase B Preliminary Design and Technology Completion	To define the project in enough detail to establish an initial baseline capable of meeting mission needs. Develop system structure end product (and enabling product) requirements and generate a preliminary design for each system structure end product.	End products in the form of mock-ups, trade study results, specification and interface documents, and prototypes
Implementation	**Phase C** Final Design and Fabrication	To complete the detailed design of the system (and its associated subsystems, including its operations systems), fabricate hardware, and code software. Generate final designs for each system structure end product.	End product detailed designs, end product component fabrication, and software development
	Phase D System Assembly, Integration and Test, Launch	To assemble and integrate the system (hardware, software, and humans), meanwhile developing confidence that it is able to meet the system requirements. Launch and prepare for operations. Perform system end product implementation, assembly, integration and test, and transition to use.	Operations-ready system end product with supporting related enabling products
	Phase E Operations and Sustainment	To conduct the mission and meet the initially identified need and maintain support for that need. Implement the mission operations plan.	Desired system
	Phase F Closeout	To implement the systems decommissioning/disposal plan developed in Phase E and perform analyses of the returned data and any returned samples.	Product closeout

Hirshorn, S. R., Voss, L. D., & Bromley, L. K. (2017). NASA systems engineering handbook (No. HQ-E-DAA-TN38707). Chicago.

Which team do you think developed each concept?

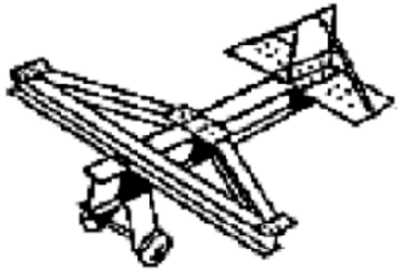
Clockwise, from top

Weights

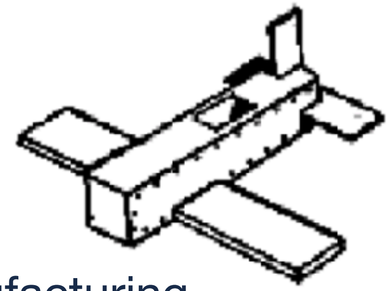


Controls

Structures



Aerodynamics



Manufacturing

Noise

Propulsion

The system designer must find the best compromise across all disciplines

Systems thinking

Systems thinking is a holistic mindset in systems engineering considering the entire system, emphasizing all of its components and their interactions.

Recall: Systems engineering aims to ensure that the system that we design, realize, and operate **meets stakeholder needs and performs the intended function**.

When focusing on a smaller piece of the puzzle (specific subsystem and/or discipline), we need to keep an eye on the **big picture.**

Concretely, systems thinking means understanding:

- Interactions between components
- Stakeholder needs
- Larger iterative process
- The entire lifecycle

Benefits of applying systems thinking:

- Fewer iterations
- Reduced late-scale, costly design changes
- Better performance, efficiency, and reliability
- More innovative designs

Nomenclature: Types of design activities

- **Selection design:** selecting from a list of options
- **Configuration design:** determining how to assemble designed components

Discrete design choices

- **Parametric design:** selecting values of identified parameters

Continuous design choices

- **Original design:** developing a process, assembly, or component not in existence
  - Redesign
  - Variant design

Combination of the above three types

Ullman, D. (2009). The mechanical design process. 4th edition. McGraw hill.

Pugh charts

Example of a Pugh chart (illustrative/notional purposes only)

Criteria	Baseline	Blended-Wing Body	Conventional Design with Winglets	Electric Propulsion
Aerodynamic Efficiency	0	+	+	–
Weight	0	+	0	–
Fuel Efficiency	0	+	+	++
Manufacturing Cost	0	–	0	–
Maintenance Complexity	0	–	0	–
Noise Emissions	0	+	+	++
Environmental Impact	0	+	+	++
Technology Readiness	0	–	+	––

Pugh charts

- Make the decision making process more systematic; but NOT more automated
- Help to quickly eliminate weak ideas
- Provide a visual comparison of concepts' pros and cons
- Encourage discussion and teamwork
- Always require interpretation and critical thought (do NOT simply sum the numbers)
- Can only be as good as the list of criteria
- Often cause a tendency to give more equal weight to the criteria than is appropriate

Analytical hierarchy process (AHP)

- Analytic Hierarchy Process is a systematic method that can be applied to trade studies.
- Benefit: The need to select from several competing alternatives is transformed into a simpler problem of defining criteria and weights, and then performing pairwise comparisons.
- The result is a quantitative scoring of each design choice for each criterion.
- Using an eigenvalue decomposition, it is possible to assess consistency.

Stability: Principal Eigenvalue = 4.1259

	Conventional	Canard	Tandem	Flying Wing	Principal Eigenvector
Conventional	1	7	4	9	0.9330
Canard	1/7	1	1/3	3	0.1434
Tandem	1/4	3	1	5	0.3227
Flying Wing	1/9	1/3	1/5	1	0.0691

Consistency index

$$CI = \frac{\lambda - n}{n - 1}$$

Agenda

System design

Aircraft design process

Conceptual design

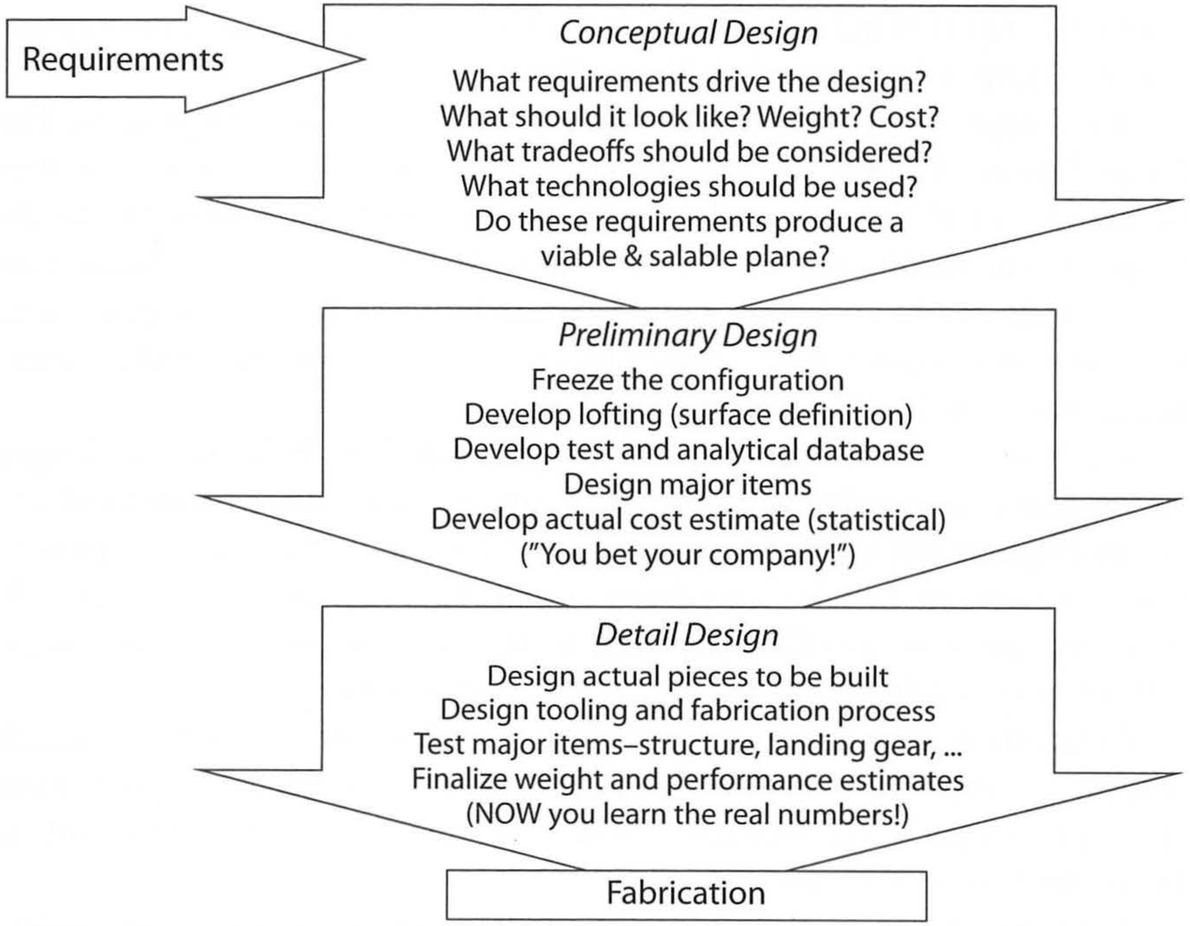
Properties of the aircraft process

The aircraft design process follows the general top-down system design approach

Therefore, it is also relies heavily on:

- Initial assumptions
- Guidance and data from existing solutions (i.e., designs)
- Iteration and refinement

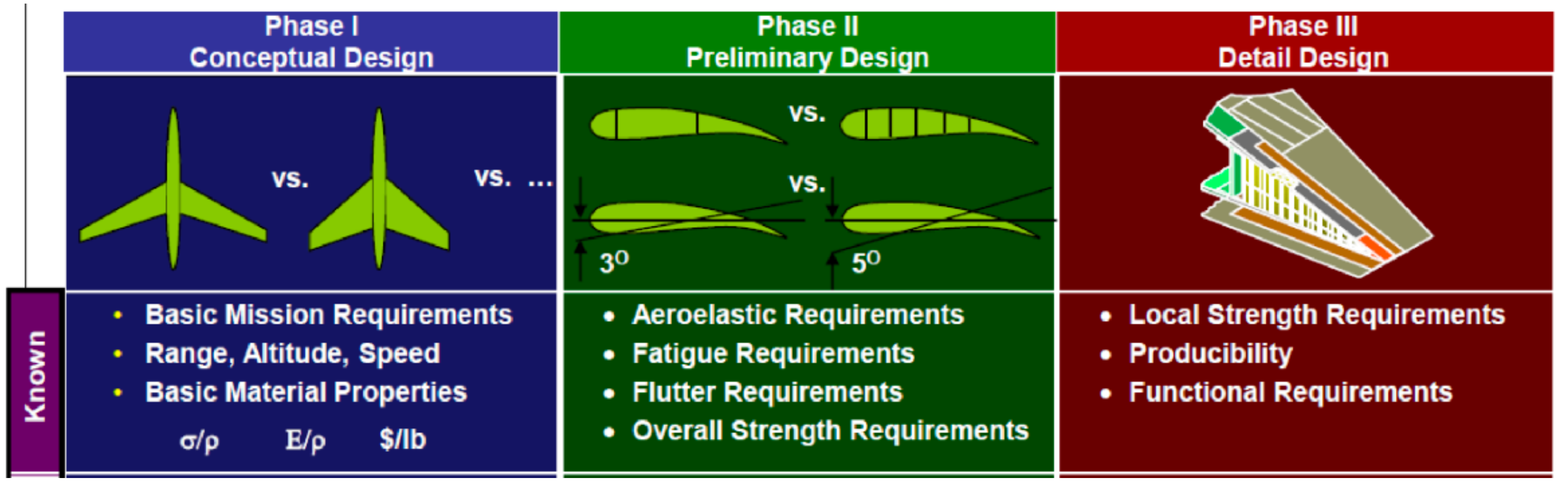
The three stages of the aircraft design process



Raymer, D. (2012). Aircraft design: a conceptual approach.
American Institute of Aeronautics and Astronautics, Inc.

Conceptual design

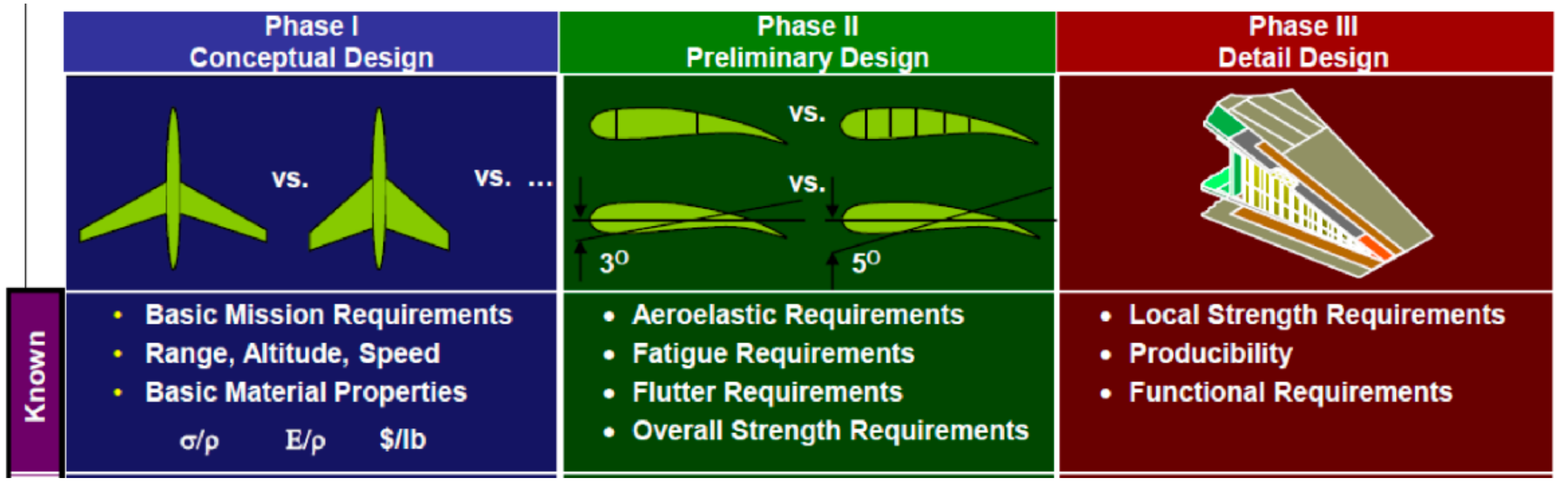
- Consider a range of configurations, perform trade studies on the design and requirements, and select a single concept.
- Determine the configuration, basic sizing and weight, and performance.
- Geometry output: overall wing/tail geometry, fuselage shape, and internal layout
- Fluid process where multiple concepts may be simultaneously studied.
- Goal: requirements, concept, and sizing that are all compatible with each other.



Credit: L. Nicolai

Preliminary design

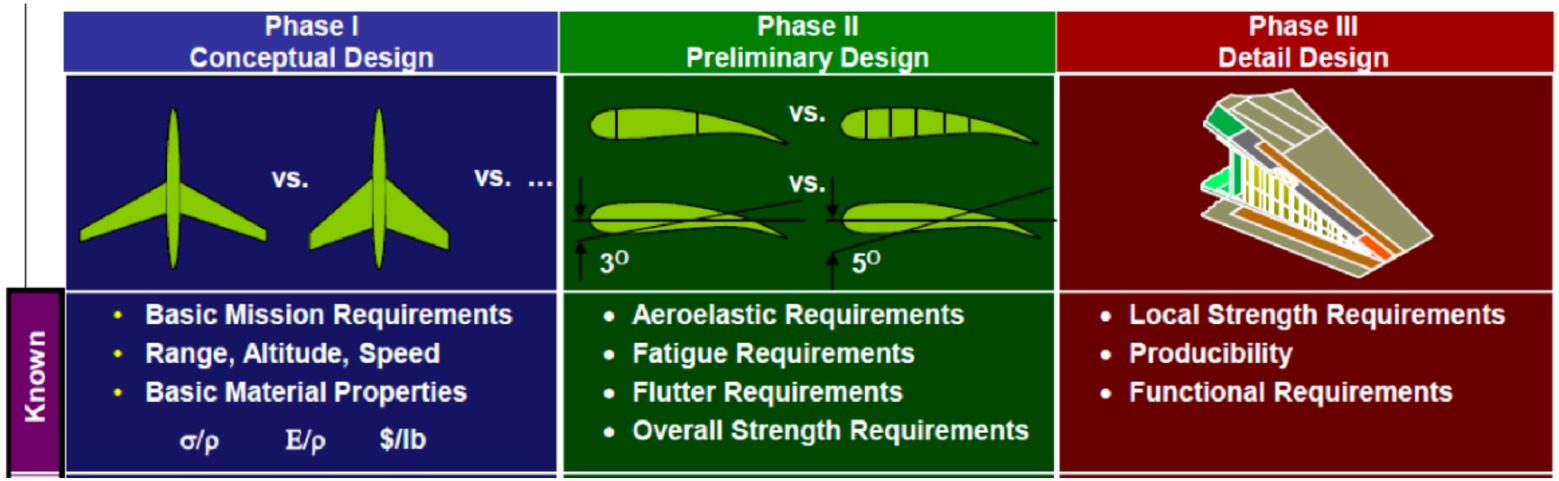
- No more major changes—configuration arrangement fixed. Use a fairly detailed CAD mockup.
- Testing in individual disciplines—aerodynamics, propulsion, structures, and stability & control.
- Example: perform structural sizing of the spar using analysis tools to ensure the thicknesses are large enough to carry the loads, but not with enough detail that the resulting geometry is actually buildable (ignore fasteners, cutouts, and so on).
- Goal: prepare for detail design—full-scale development.



Credit: L. Nicolai

Detail design

- Full-scale development—actual parts are designed.
- This step involves the vast majority of the engineers.
- Example: structural analysis in preliminary design involved the full wingbox, but here the analysis is of a single rib.
- Goal: final design that will be built.



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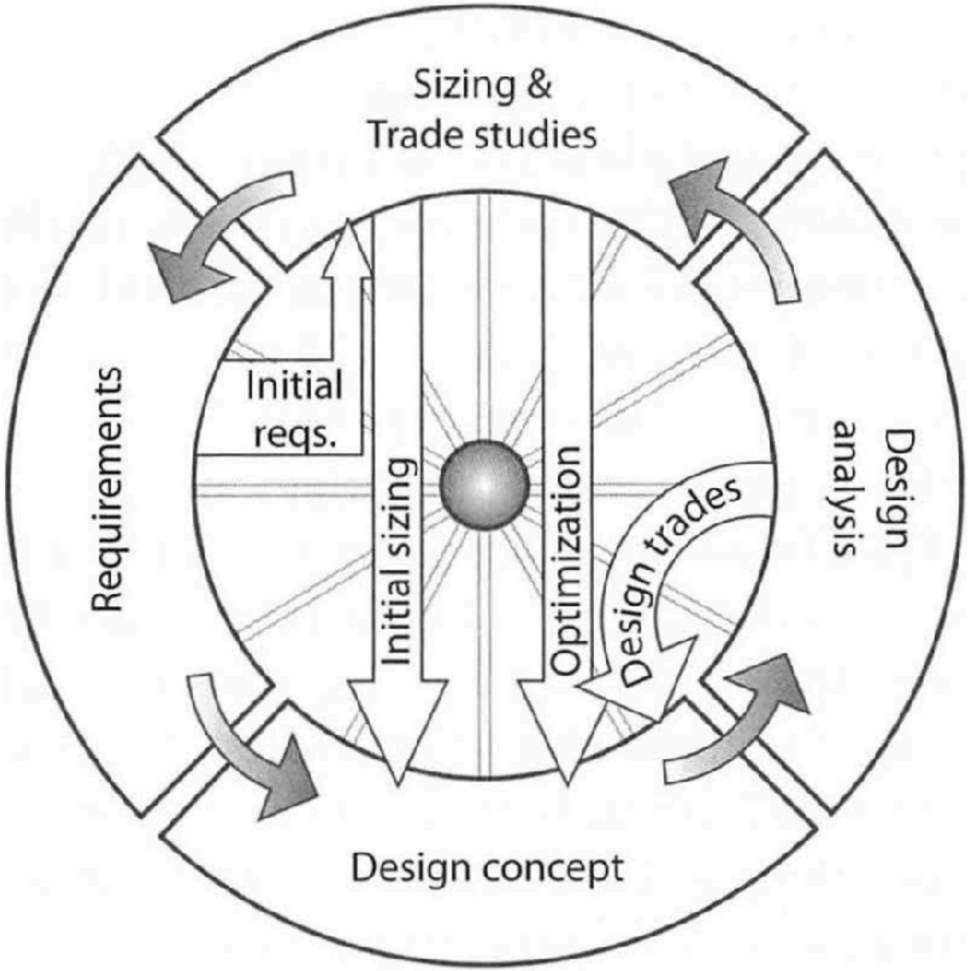
The aircraft conceptual design wheel



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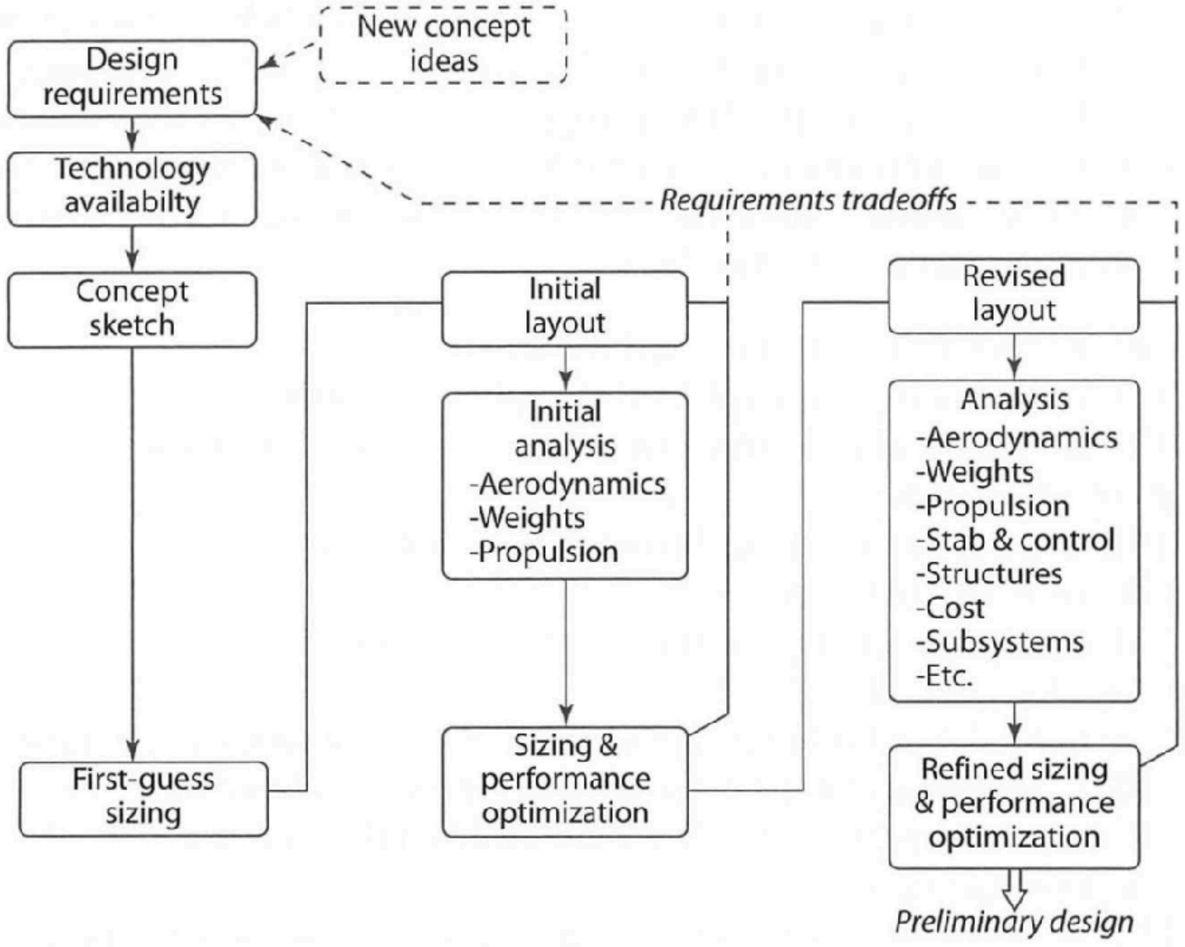
An example of a conceptual design workflow



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The design process starts with requirements

- Requirements are based on: the aircraft's purpose and operation, customer/operator assumptions, development time frame, risk tolerance.
- Company typically sets the requirements based on market studies, customer input, and the competition—moreso in civil aviation, less on the military side (even then, companies heavily involved in RFP development).
- Maximum takeoff gross weight (MTOW) is largely determined by payload weight and aircraft mission range requirements.
- Engine size and wing size (area) are largely determined by requirements on maximum speed, stall speed, rate of climb, takeoff distance, and other performance numbers.

Airworthiness standards

- Federal Aviation Regulations (FARs) are rules from the Federal Aviation Administration (FAA) for safe aviation in the US.
- They deal with performance, stability, controllability, takeoff, landing, handling qualities, etc.
- Part of Title 14 of the Code of Federal Regulations (CFR).
- Regulations grouped by aircraft type and weight class:
    - 14 CFR Part 23: Normal category airplanes (<12,500 lb; 19 passengers)
    - 14 CFR Part 25: Transport category airplanes
    - 14 CFR Part 27: Normal category rotorcraft (<17,000 lb; 9 passengers)
    - 14 CFR Part 29: Transport category rotorcraft

Example of FARs

§23.2100 Weight and center of gravity.

- (a) The applicant must determine limits for weights and centers of gravity that provide for the safe operation of the airplane.
- (b) The applicant must comply with each requirement of this subpart at critical combinations of weight and center of gravity within the airplane's range of loading conditions using tolerances acceptable to the Administrator.
- (c) The condition of the airplane at the time of determining its empty weight and center of gravity must be well defined and easily repeatable.

Types of requirements

Mission requirements:

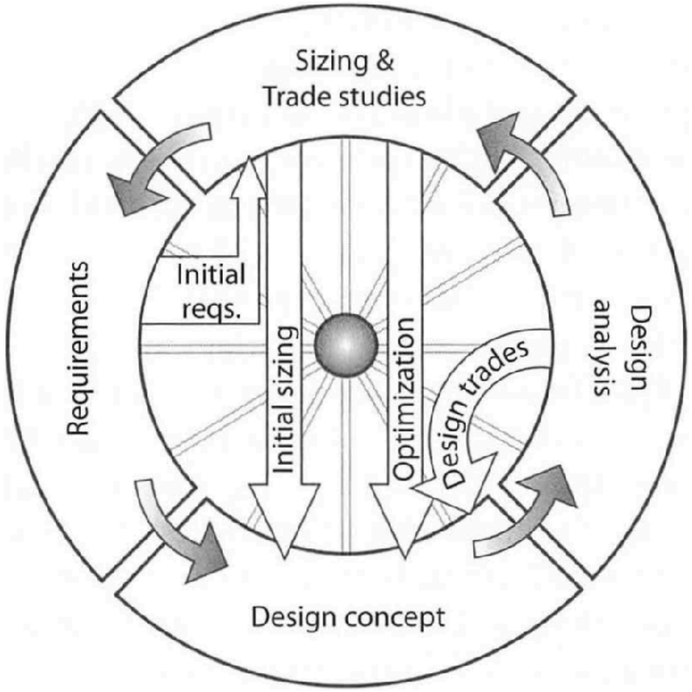
- Range
- Endurance and/or loiter duration
- Purpose: e.g., commercial transport; general aviation; intelligence, surveillance, reconnaissance (ISR);
- trainer; air-to-air fighter; bomber
- Payload: e.g., passengers, cargo, sensors
- Speed: e.g., cruise, maximum, loiter, landing
- Field length: e.g., VTOL (vertical takeoff and landing), STOL (short), CTOL (conventional)
- Signature level: e.g., radar cross section (RCS), infrared (IR), visual, acoustic

Cost requirements:

- Development cost
- Acquisition cost
- Operation and support (O&S cost)
- Life cycle cost (LCC): development, acquisition, O&S combined

The upcoming lectures will cover the remaining parts of conceptual design

- Sizing
- Configuration
- Aerodynamics
- Propulsion
- Weights and balance
- Structures and materials



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American Institute of Aeronautics and Astronautics, Inc.

Summary and key takeaways

Successful aircraft design requires applying systems thinking—always keep the bigger picture in mind

Early decisions have the largest impact on cost, performance, and feasibility

System design is a top-down, iterative process—start from requirements; make assumptions; use existing designs; iterate

Trade studies help you structure and justify design decisions—use tools like Pugh charts to support your decisions

Conceptual design aims to converge to a configuration, sizing, and requirements that are mutually compatible

Requirements come from the mission, stakeholders, and certification considerations

- **Apply this mindset continuously throughout the quarter**

Reference materials

Useful websites

- Electric VTOL News: https://evtol.news
- Vertical Flight Society (VFS): https://vtol.org
- NASA Advanced Air Mobility (AAM) Project: https://www.nasa.gov/aeroresearch/programs/iasp/aam
- NASA Revolutionary Vertical Lift Technology (RVLT) Project: https://www.nasa.gov/aeroresearch/programs/aavp/rvlt
- NASA UAS Traffic Management (UTM) Project: https://www.nasa.gov/utm
- NASA Air Traffic Management – eXploration (ATM-X) Project: https://www.nasa.gov/aeroresearch/programs/aosp/atm-x
- NASA Transformational Tools and Technologies (TTT) Project: https://www.nasa.gov/aeroresearch/programs/tacp/ttt
- NASA Aeronautics Systems Analysis Branch: https://sacd.larc.nasa.gov/asab/asab-projects-2/
- NASA Electrified Aircraft Propulsion (EAP) Concepts: https://www1.grc.nasa.gov/aeronautics/eap/airplane-concepts/
- Google Scholar: https://scholar.google.com

10 Rules of Systems Analysis

1. There Always Is a Client. (Need to understand who the client is)
2. Your Client Does Not Understand Their Own Problem. (Part of your job is to frame the problem)
3. The Original Problem Statement is Too Specific: You Must Generalize the Problem to Give it Contextual Integrity. (The problem system often has missing information or context—treat the illness, not the symptoms)
4. The Client Does Not Understand the Concept of the Index of Performance. (You must sometimes educate the client)
5. You Are the Analyst, Not the Decision-Maker. (Provide insights/options for the client, not dictate decisions)
6. Meet the Time Deadline and the Cost Budget. (Simplify or make assumptions if necessary)
7. Take a Goal-Centered Approach to the Problem, Not a Technology-Centered or Chronological Approach. (Focus on achieving the objectives; do not get married to a specific tool, technology, etc.)
8. Nonusers Must Be Considered in the Analysis and in the Final Recommendations. (Considering communities that might be affected or the environment is crucial for success in the long term)
9. The Universal Computer Model is a Fantasy. (No single model can capture all aspects; simplifications are necessary)
10. The Role of Decision-Maker in Public Systems is Often a Confused One. (Complexity due to politics, conflicting goals)

Gibson, J. E., Scherer, W. T., & Gibson, W. F. (2007). How to do systems analysis. John Wiley & Sons.

A general decision-making process (selection design)

1. **Clarify the issue** that needs a satisfactory solution.
2. **Generate alternatives**—itemize the potential solutions for the issue.
3. **Develop criteria** as they measure a satisfactory solution for the issue.
4. **Identify criteria importance** of each criterion relative to the others.
5. **Evaluate** the value of the alternatives by comparing them to the criteria.
6. Based on the evaluation results, **decide** what to do next.

Verbatim from: Ullman, D. (2009). The mechanical design process. 4th edition. McGraw hill.

A general process for original design

1. Evaluate (or define) the requirements: customers/regulations, constraints/performance goals.
2. Understand current approaches (what's done now?).
3. Think of some possible solutions (creativity).
4. Identify a variety of possible concepts (concept generation).
5. Perform concept evaluation (analysis).
6. Select a preferred concept for development (make a decision).
7. Do the detailed design and create a prototype (analysis).
8. Test and evaluate the design (scrutiny).
9. Continually refine the design until it's a viable product.

Verbatim from: AOE 4065 - 4066 Air Vehicle Design. Professor William Mason. Virginia Tech.

Analytical hierarchy process (AHP)—worked example

Stability: principal eigenvalue = 4.1259					
	Conventional	Canard	Tandem	Flying wing	Principal eigenvector
Conventional	1	7	4	9	0.9330
Canard	1/7	1	1/3	3	0.1434
Tandem	1/4	3	1	5	0.3227
Flying wing	1/9	1/3	1/5	1	0.0691

Lift: principal eigenvalue = 4.1332					
	Conventional	Canard	Tandem	Flying wing	Principal eigenvector
Conventional	1	1/5	1/9	1	0.0782
Canard	5	1	1/5	5	0.3100
Tandem	9	5	1	9	0.9443
Flying wing	1	1/5	1/9	1	0.0782

Weight: principal eigenvalue = 4.1106					
	Conventional	Canard	Tandem	Flying wing	Principal eigenvector
Conventional	1	2	4	1/6	0.2497
Canard	1/2	1	2	1/5	0.1519
Tandem	1/4	1/2	1	1/9	0.0785
Flying wing	6	5	9	1	0.9531

Simplicity: principal eigenvalue = 4.0788					
	Conventional	Canard	Tandem	Flying wing	Principal eigenvector
Conventional	1	3	6	9	0.9076
Canard	1/3	1	3	5	0.3794
Tandem	1/6	1/3	1	3	0.1634
Flying wing	1/9	1/5	1/3	1	0.0751

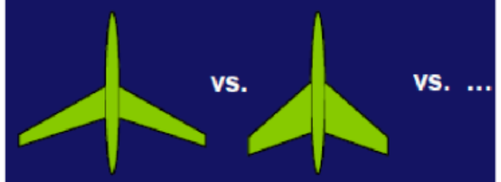
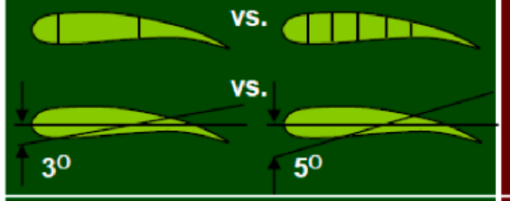
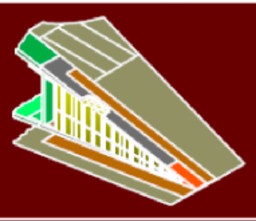
$$Stability \quad CR = \frac{\lambda - n}{RI(n-1)} = \frac{4.1259 - 4}{0.9 \times 3} = 4.66\%$$

$$Lift \quad CR = \frac{\lambda - n}{RI(n-1)} = \frac{4.1332 - 4}{0.9 \times 3} = 4.93\%$$

$$Weight \quad CR = \frac{\lambda - n}{RI(n-1)} = \frac{4.1106 - 4}{0.9 \times 3} = 4.10\%$$

$$Simplicity \quad CR = \frac{\lambda - n}{RI(n-1)} = \frac{4.0788 - 4}{0.9 \times 3} = 2.92\%$$

The three stages of the aircraft design process

	Phase I Conceptual Design	Phase II Preliminary Design	Phase III Detail Design
			
Known	• Basic Mission Requirements • Range, Altitude, Speed • Basic Material Properties σ/ρ E/ρ $/lb	• Aeroelastic Requirements • Fatigue Requirements • Flutter Requirements • Overall Strength Requirements	• Local Strength Requirements • Producibility • Functional Requirements
Results	Geometry: Airfoil Type, AR, t/c, λ, Δ Design Objectives: Drag Targets, Weight Targets, Cost Targets, Etc.	• Basic Internal Arrangement • Complete External Configuration – *Camber, Twist Distributions* – *Local Flow Problems Solved* • Major Loads, Stresses, Deflections • Etc.	• Detail Design – *Mechanisms* – *Joints, Fittings & Attachments* • Design Refinements as Results of Test
Output	Feasible Design	Mature Design	Shop Drawings
TRL	2-3	4-5	6-7

Credit: L. Nicolai